ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

# BASIC PRINCIPLES OF COMPUTER PROGRAMMING AND COMPUTER LITERACY

## PAPER 2 · PRACTICE D

**MARKS: 100** **TIME: 3 hours**

**NC(V) Programming and Robotics, Level 2. Laid out like the DHET second paper (practical, 100 marks): a Scratch recycling-points app with an IPO chart, a Python turtle drawing of coloured squares, and a Python school-transport fee program with a function, selection, a discount, a loop and a global total.**

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

## INSTRUCTIONS AND INFORMATION

1. This question paper consists of THREE questions. Answer ALL the questions.
2. This is a practical paper. Use Scratch for QUESTION 1 and a Python IDE (Python 3) for QUESTIONS 2 and 3. If you do not have a computer, write the programs neatly in your answer book.
3. Save each program with the question number and your name or examination number. Add your name or examination number as a comment on the first line of each Python program.
4. Save your work regularly. Programs are not marked strictly on syntax.

## QUESTION 1

A school runs a recycling competition. Learners bring plastic, glass and paper, and earn points: 3 points per kg of plastic, 2 points per kg of glass and 1 point per kg of paper. A learner with 50 points or more earns a GOLD badge, 25 to 49 points a SILVER badge, and fewer than 25 points a BRONZE badge. Create a Scratch app in which the sprite asks for the mass of each material, works out the points and says the badge.

| INPUT | PROCESS | OUTPUT |
|---|---|---|
| kg of plastic<br>kg of glass<br>kg of paper | Declare all variables and set them to 0<br>Ask for the three masses<br>points = plastic × 3 + glass × 2 + paper × 1<br>If points ≥ 50: badge = "GOLD"<br>Else if points ≥ 25: badge = "SILVER"<br>Else: badge = "BRONZE" | points<br>badge |

**1.1** Open Scratch, create a new project and name it Question1_YourName. (1)

**1.2** Choose a sprite to act as the competition judge. (1)

**1.3** Choose a suitable backdrop (for example a school or outdoors). (1)

**1.4** Declare all the variables and give them their starting values. (8)

**1.5** Ask for the mass of plastic, glass and paper, and store each answer. (6)

**1.6** Calculate the points. (4)

**1.7** Use if … then … else blocks to choose the badge. (8)

**1.8** Display the points and the badge. (3)

**1.9** Test the program with these values and write down what the sprite says: 10 kg plastic, 5 kg glass, 8 kg paper; 20 kg plastic, 0 kg glass, 0 kg paper; 2 kg plastic, 3 kg glass, 4 kg paper. (3)

**[35]**

## QUESTION 2

Write a Python program that uses the turtle graphics library to draw a row of EIGHT squares, side by side, each with sides of 40 pixels. The squares must be filled in alternating colours: red, blue, red, blue …



**FIGURE 2.1: The row of squares**

**2.1** Create a new file called Question2_YourName.py and add your name or examination number as a comment on the first line. (1)

**2.2** Import the turtle module and create a turtle object named builder. (2)

**2.3** Set the drawing speed to 0. (2)

**2.4** Set the thickness of the lines to 2 pixels. (2)

**2.5** Use a loop that draws the square 8 times, moving the turtle 40 pixels to the right before each new square. (5)

**2.6** Use a second (inner) loop to draw each square: 4 sides of 40 pixels with a 90° turn after each side. (5)

**2.7** Fill the squares in alternating colours (red for the 1st, 3rd, 5th … square and blue for the others), using an if statement. (5)

**[22]**

## QUESTION 3

A school organises bus transport for learners. Learners who live less than 5 km from the school walk and pay nothing. Learners who live from 5 km up to 15 km pay R150 per month, and learners who live more than 15 km away pay R250 per month. A learner who has a brother or sister on the bus gets a 20% discount. The school wants a Python program that registers learners one by one and shows the total monthly fees at the end.

The following steps show how the program should be used:

```
Learner name: Ayanda
Distance from school (km): 8
Sibling on the bus? (yes/no): no
Ayanda pays R150.00 per month
Register another learner? (yes/no): yes
Learner name: Kabelo
Distance from school (km): 20
Sibling on the bus? (yes/no): yes
Kabelo pays R200.00 per month (20% sibling discount)
Register another learner? (yes/no): no
Total monthly fees: R350.00
```

Follow the instructions below to create the application:

**3.1** Create a new Python file called Question3_YourName.py. (1)

**3.2** Declare a global variable named total_fees and set it to 0. (2)

| | | |
|---|---|---|
| **3.3** | Create a function named register_learner() with no parameters. Inside it, ask for the learner's name, the distance from school and whether a sibling is on the bus. | (6) |
| **3.4** | Use if, elif and else to choose the monthly fee from the distance. | (8) |
| **3.5** | Give a 20% discount when the learner has a sibling on the bus. | (4) |
| **3.6** | Display the fee for the learner. | (4) |
| **3.7** | Use a loop that keeps registering learners until the answer is "no". | (7) |
| **3.8** | Add each learner's fee to total_fees. | (4) |
| **3.9** | Call the function inside the loop and display the total monthly fees at the end. | (4) |
| **3.10** | Test the program with the data shown in the example above. | (2) |
| **3.11** | Save the program and print (or neatly present) the code. | (1) |
| | | **[43]** |
| | | **TOTAL: 100** |

**END OF QUESTION PAPER**